VIRAT AGNIHOTRI

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PROFESSIONAL SUMMARY

Second-year Computer Science undergraduate with hands-on experience building full-stack and applied AI projects in scientific simulation, computer vision, speech analytics, and retrieval-augmented generation. Experience with Python/FastAPI, React, REST APIs, numerical methods, and pretrained model integration.

TECHNICAL SKILLS

Languages: Python, C++, JavaScript, TypeScript, HTML, CSS, SQL (basic)
Web & Data: React, FastAPI, REST APIs, PostgreSQL, SQLite, SQLAlchemy, Supabase
AI & Retrieval: PyTorch, Hugging Face Transformers, Whisper, Wav2Vec2, MiniLM, RAG, FAISS
Vision & Scientific: OpenCV, SIFT, FLANN/KNN, RANSAC, optical flow, librosa, NumPy, SciPy
Other: Three.js, React Three Fiber, Recharts, Git, pytest

TECHNICAL PROJECTS

Rocket Mission Simulator | Python, SciPy, FastAPI, React, TypeScript

- Built a 2D Earth-centered rocket ascent simulator modeling gravity, atmospheric drag, propellant depletion, and pitch guidance.
- Used SciPy solve_ivp for numerical integration and differential evolution for constrained trajectory optimization.
- Created 3D trajectory playback, telemetry charts, SQLite mission storage, and CSV export; backend validated with 43 passing tests.

Lunar Vision | Python, OpenCV, FastAPI, React

- Built a lunar image-registration workflow using Gaussian/CLAHE preprocessing and SIFT feature detection.
- Applied FLANN/KNN matching, Lowe ratio filtering, and RANSAC homography to align images and reject outliers.
- Used Lucas–Kanade optical flow for point refinement and displayed overlays, difference maps, keypoints, and inlier statistics.

Knowledge Vault AI | React, FastAPI, PostgreSQL, SQLAlchemy, FAISS

- Built a JWT-authenticated knowledge workspace with editable pages, autosave, favorites, trash, and persistent AI conversations.
- Implemented PDF ingestion with PyMuPDF, semantic chunking, and MiniLM embeddings for document-aware retrieval.
- Integrated FAISS similarity search and Qwen responses; used PostgreSQL/SQLAlchemy for persistence and Supabase for file storage.

Tone Analyser | Python, PyTorch, Whisper, Wav2Vec2, librosa

- Integrated pretrained speech transcription and emotion models; extracted pitch, RMS energy, zero-crossing rate, and speech activity.
- Developed a heuristic vocal-stress index using normalized acoustic features and weighted emotion scores.
- Built session/lap analytics, RAG-assisted racing insights, and React/Recharts dashboards.

DocIntel AI | Python, FastAPI, React, FAISS, Qwen

- Built a PDF question-answering workflow with PyMuPDF extraction, semantic text chunking, and MiniLM embeddings.
- Used FAISS top-k retrieval to supply relevant document context for Qwen-generated answers.

EDUCATION

ABES Institute of Technology, Ghaziabad | B.Tech, Computer Science & Engineering (Core) | Year 2 | CGPA: 8.1/10
St. Joseph's School, Kanpur | Class XII: 76% | Class X: 89%